Multilevel Analysis of Individual Heterogeneity and Discriminatory Accuracy (MAIHDA)

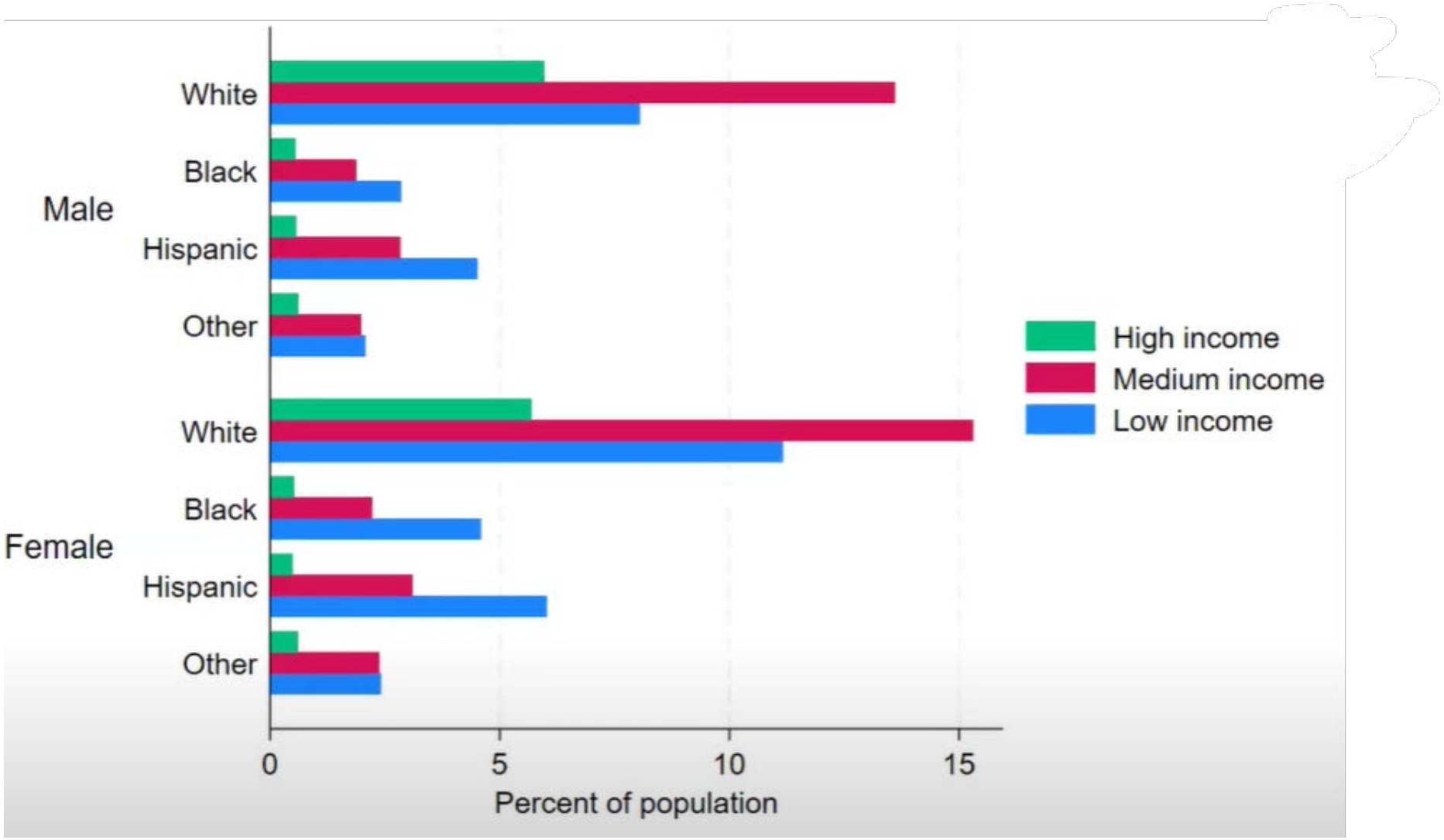
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What is MAIHDA

- New approach exploring sociodemographic inequalities. Focuses on **intersections** of characteristics
- Originally proposed in social epidemiology by Evans et al. (2018) looking at how BMI differs in gender, race, education, income, and age.
- MAIHDA is slowly creeping into studies dealing with disparities in pharmacoepidemiology including:
  - Drug use, self-medication, medication sharing, adherence compliance, adverse reactions, etc.

What is Intersectionality

- Development of MAIHDA was motivated by intersectionality theory.
- Intersectionality is the study of how various systems of oppression / privilege / power intersect and create inequalities across social identities.
- Key aim is to make visible the unique lived experiences and outcomes at different intersections, especially those of the multiply marginalized.


Male
White
Black
Hispanic
Other
Female
White
Black
Hispanic
Other
High income
Medium income
Low income
0
5
10
15
Percent of population

Conventional Regression with Interactions

- Traditional approach is a regression model with interactions
  - This can get out of hand quickly: might start with sex*race but if we want to include age, income, and other sociodemographic variables we end up with lots of interactions.

- Faced with these challenges we try to simplify to decreasing number of interactions

MAIHDA Model

- Two-Level multilevel regression:
  - Level 1: Individuals
  - Level 2: intersections of individuals' social identities

- Focus on mapping means and mean differences between intersections

MAIHDA Model

- Allows us to answer these research questions:
  - How big are intersectional inequalities
  - Which intersections are most advantaged / disadvantaged
  - Which intersections deviate most/least from expectations

- MAIHDA does not identify or test the role of systems of oppression. Still relies on risk factors and not causes

MAIHDA Model

- Model 1: Simple Intersectional Model

$$y_{ij} = \beta_0 + u_j + e_{ij}$$

where $u_j \sim N\left(0, \sigma_u^2\right)$ and $e_{ij} \sim N\left(0, \sigma_e^2\right)$.

- Precision Weighted Grand Mean (PWGM) is calculated by calculating predicted means which are reliability weighted average of the observed sample mean

$$\widehat{\beta}_0 = \frac{\sum_{j=1}^{J} w_j \bar{y}_j}{\sum_{j=1}^{J} w_j}, w_j = \frac{1}{\widehat{\sigma}_u^2 + \frac{\widehat{\sigma}_e^2}{n_j}}$$

MAIHDA Model

- Model 2: Linear Models with Main Effects

$$y_{ij} = \beta_0 + \beta_1 x_{1j} + \cdots + \beta_p x_{pj} + u_j + e_{ij}$$

$$u_j \sim N\left(0, \sigma_u^2\right)$$

$$e_{ij} \sim N\left(0, \sigma_e^2\right)$$

Example: HbA1c Data

- Simulated Data which is based on Wave 2 of the National Epidemiologic Survey on Alcohol and Related Conditions (NESARC).
- Looking to see if HbA1c levels are different among groups of interest:
  - Sex (Male, Female)
  - Race (White Non-Hispanic, Black Non-Hispanic, Hispanic/Latino)
  - Education (< HS, HS, Some College, College or More)
  - Age (18-29, 30-44, 45-59, 60+)
  - Income (Low, Low-Middle, High-Middle, High).
- N= 33,000

Example: HbA1c Data

Table 1
Descriptive statistics of individual observations (N = 33,000) and intersectional strata (n = 384) in the simulated data.

Sample Statistics	N		%			
Total	33,000		100			
Sex						
Male	13,931		42.2			
Female	19,069		57.8			
Race/Ethnicity						
White Non-Hispanic	20,075		60.8			
Black Non-Hispanic	6572		19.9			
Hispanic/Latino	6353		19.3			
Education						
Less than high school	5272		16.0			
Completed high school	9097		27.6			
Some college no degree	7080		21.5			
College degree or more	11,551		35.0			
Income (% Poverty Threshold)						
Low income (<100%)	7712		23.4			
Low-middle income (100%–199%)	9198		27.9			
High-middle income (200%–399%)	9605		29.1			
High income (400% or more)	6485		19.7			
Age						
18–29 years	4657		14.1			
30–44 years	10,048		30.5			
45–59 years	9213		27.9			
60+ years	9082		27.5			
HbA1c	N	Mean	Median	SD	Min	Max
Individual values	33,000	40.2	39.8	9.9	10.3	101.2
Observed stratum means	384	40.8	40.3	3.9	23.5	53.8
Predicted stratum means	384	40.8	40.5	2.7	34.2	48.1

Notes: The mean of the 33,000 individual observations of HbA1c is the "sample mean." The mean of the 384 observed stratum means is the "grand mean." The mean of the 384 predicted stratum means from the MAIHDA model is the "precision-weighted grand mean."

Example: HbA1c Data

- Good idea to see how many stratums have a small sample size

Table 2. Sample Size of Simulated Intersectional Social Strata, defined by respondent sex, race/ethnicity, education, income, and age (n=384).

Sample Size Per Stratum	Number of Strata	% of Strata
100 or More	107	27.9
50 or More	202	52.6
30 or More	267	69.5
20 or More	311	81.0
10 or More	347	90.4
Less than 10	37	9.6

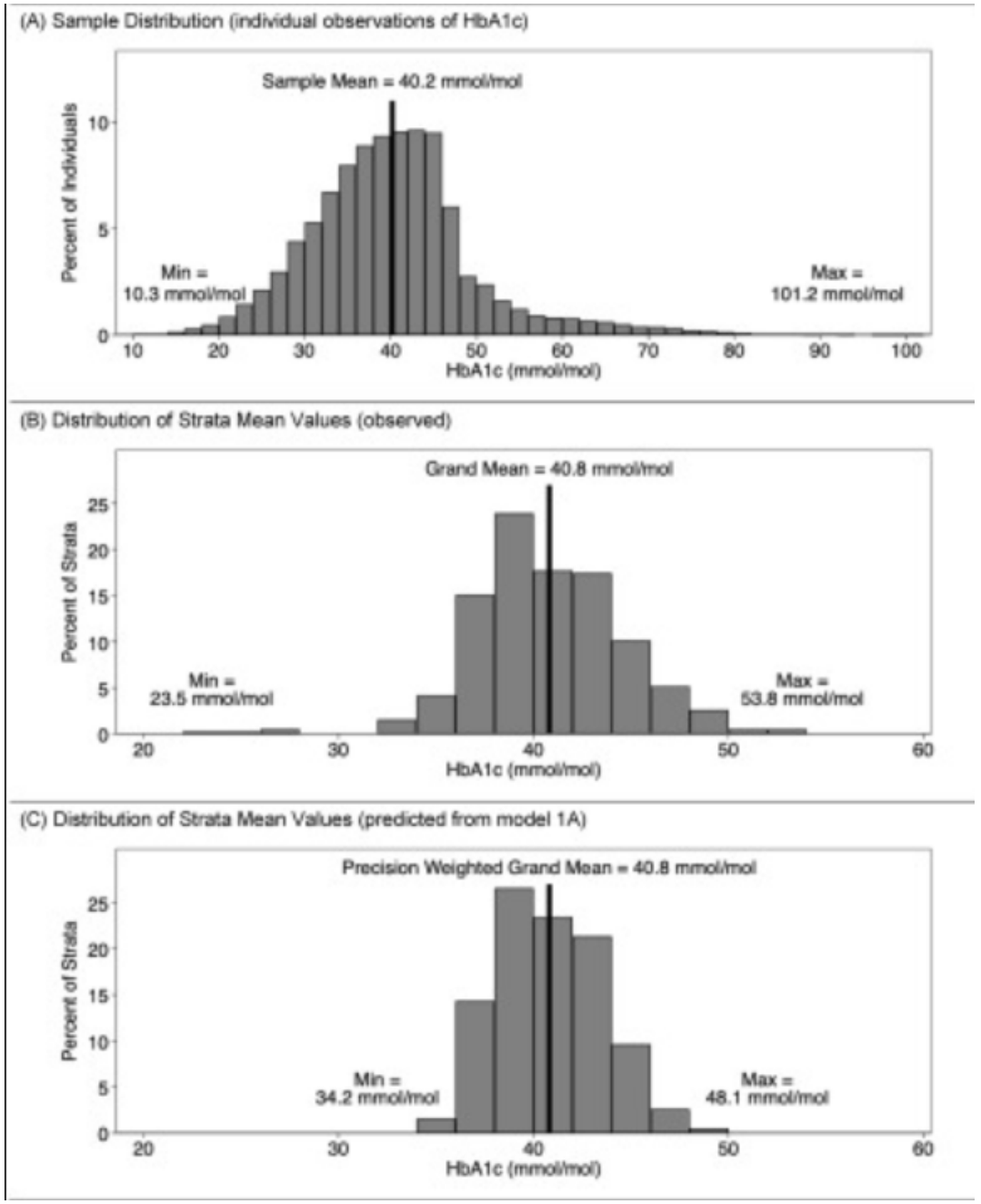

(A) Sample Distribution (individual observations of HbA1c)
Sample Mean = 40.2 mmol/mol
Percent of Individuals
Min = 10.3 mmol/mol
Max = 101.2 mmol/mol
HbA1c (mmol/mol)
(B) Distribution of Strata Mean Values (observed)
Grand Mean = 40.8 mmol/mol
Percent of Strata
Min = 23.5 mmol/mol
Max = 53.8 mmol/mol
HbA1c (mmol/mol)
(C) Distribution of Strata Mean Values (predicted from model 1A)
Precision Weighted Grand Mean = 40.8 mmol/mol
Percent of Strata
Min = 34.2 mmol/mol
Max = 48.1 mmol/mol
HbA1c (mmol/mol)

Table 3
Parameter estimates for linear models of HbA1c (mmol/mol) and logistic models of diabetes (HbA1c $\geq$ 48 mmol/mol).

	Linear Model 1A		Linear Model 1B		Logistic Model 2A		Logistic Model 2B	
	Estimate	[95% CI]	Estimate	[95% CI]	OR	[95% CI]	OR	[95% CI]
Fixed Effects: Regression Coefficients								
Intercept	40.79	[40.44, 41.13]	38.03	[37.45, 38.62]	0.16	[0.15, 0.17]	0.09	[0.07, 0.10]
Sex								
Male (Ref)			–				–	
Female			−0.52	[-0.83, −0.21]			0.90	[0.83, 0.98]
Race/Ethnicity								
White Non-Hispanic (Ref)			–				–	
Black Non-Hispanic			4.45	[4.08, 4.82]			2.36	[2.15, 2.59]
Hispanic/Latino			1.04	[0.66, 1.42]			1.19	[1.07, 1.32]
Education								
Less than high school (Ref)			–				–	
Completed high school			−0.54	[-1.01, −0.07]			0.94	[0.83, 1.06]
Some college no degree			−0.55	[-1.03, −0.06]			0.89	[0.79, 1.02]
College degree or more			−1.13	[-1.61, −0.65]			0.81	[0.71, 0.92]
Income (% Poverty Threshold)								
Low (<100%) (Ref)			–				–	
Low-middle (100%–199%)			0.18	[-0.23, 0.60]			1.09	[0.98, 1.22]
High-middle (200%–399%)			−0.30	[-0.73, 0.12]			0.96	[0.86, 1.08]
High (400% or more)			−1.33	[-1.82, −0.83]			0.79	[0.69, 0.91]
Age								
18–29 years (Ref)			–				–	
30–44 years			1.10	[0.65, 1.56]			1.35	[1.18, 1.55]
45–59 years			1.81	[1.35, 2.27]			1.55	[1.35, 1.77]
60+ years			5.50	[5.03, 5.98]			3.06	[2.68, 3.49]
Random Effects: Variances								
Stratum-Level	9.33	[7.82, 11.14]	0.80	[0.55, 1.15]	0.39	[0.32, 0.49]	0.03	[0.02, 0.06]
Individual-Level	90.26	[88.89, 91.66]	90.27	[88.89, 91.66]	–			
Summary Statistics								
Variance Partition Coefficient (VPC)	9.4%		0.9%		10.7%		0.9%	
Proportional Change in Variance (PCV)			91.4%				92.2%	
Area Under Receiver Operating Characteristic Curve (AUC)					0.68		0.67	

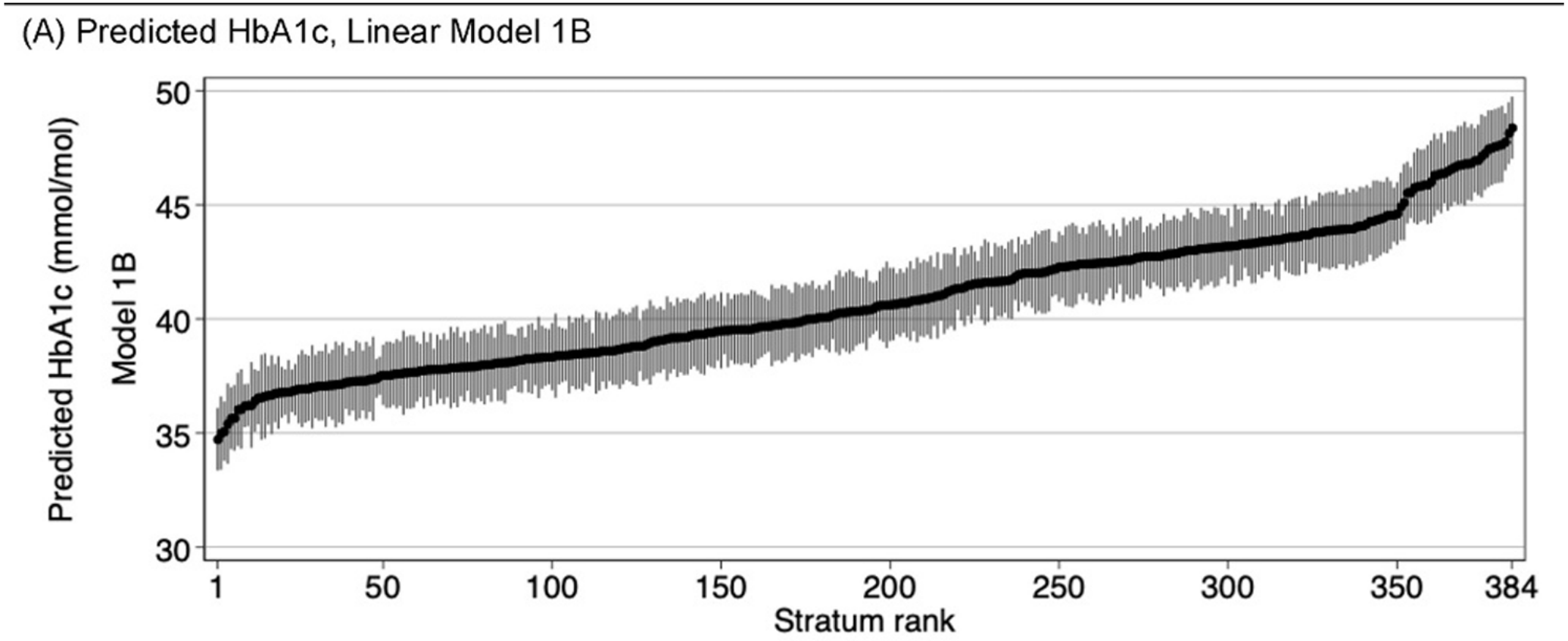
Notes: MLE estimation used for all models shown. All values are derived from simulated data and are for demonstration purposes only. 95% CIs shown in parentheses. VPC for logistic models are calculated using the latent response approach (σ_e^2 is set equal to the variance of the standard logistic distribution $\frac{\pi^2}{3} \approx 3.29$ where π here denotes the mathematical constant 3.14159).

Example: HbA1c Data

Table 4
Inspection of five highest and lowest ranked strata for predicted mean HbA1c (Model 1B) and percentage diabetic (model 2B).

Predicted Mean HbA1c (Model 1B)									
Rank	Stratum	Sex	Race	Education	Income	Age	n	Predicted Mean HbA1c	Approximate 95% CI
5 Lowest									
1	11441	Male	White	College plus	High	18–29	111	34.7	[33.4, 36.1]
2	21341	Female	White	Some college	High	18–29	43	35.0	[33.4, 36.6]
3	21331	Female	White	Some college	Mid-high	18–29	136	35.1	[33.8, 36.4]
4	21241	Female	White	High school	High	18–29	15	35.4	[33.7, 37.2]
5	21441	Female	White	College plus	High	18–29	114	35.7	[34.3, 37.0]
5 Highest									
380	12324	Male	Black	Some college	Low-mid	60+	33	47.6	[46.0, 49.3]
381	12134	Male	Black	< High school	Mid-high	60+	29	47.7	[46.0, 49.3]
382	22214	Female	Black	High school	Low	60+	157	47.8	[46.5, 49.0]
383	22124	Female	Black	< High school	Low-mid	60+	114	48.1	[46.8, 49.5]
384	12114	Male	Black	< High school	Low	60+	112	48.4	[47.0, 49.7]

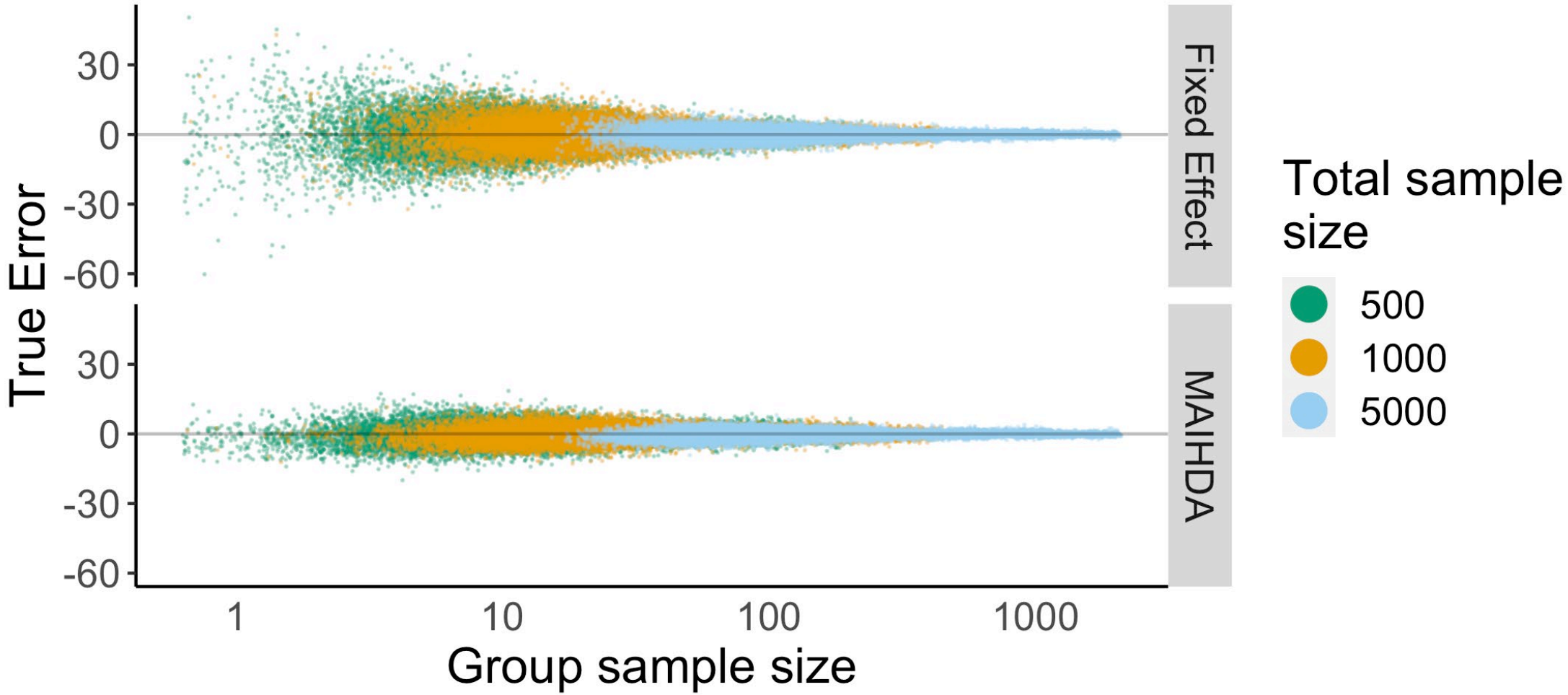
Example: HbA1c Data



MAIHDA vs Interaction Models

- MAIHDA:
  - Simplifies the inclusion of additional intersectional terms
  - Improves accuracy for smaller strata
  - More power

- Van Dusen et al. (2024) found that MAIHDA models could generate more accurate estimates of outcomes for groups with a minimum sample size of 10 than fixed-effect models could do with a minimum sample size of 20.


True Error
30
0
-30
-60
Fixed Effect
MAIHDA
1
10
100
1000
Group sample size
Total sample size
500
1000
5000

Limitations

- Theoretical:
  - Must pair it with appropriate theory when creating categories and including the correct sociodemographic variables of interest
- Technical:
  - Need at least 20 strata
  - Recommend using Bayesian statistics

References

- Van Dusen, B., Cian, H., Nissen, J., Arellano, L., & Woods, A. D. (2024). Comparing the Efficacy of Fixed-Effects and MAIHDA Models in Predicting Outcomes for Intersectional Social Strata. Sociology of Education, 97(4), 342-362. https://doi.org/10.1177/00380407241254092 (Original work published 2024)
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